

AR(1) SVP models in `svpChange2`

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This note describes every AR(1) interface used by SVP in `svpChange2`: the generator, robust estimation of ρ , the exact fixed-variance and profiled tests, the functional-pruning implementation, and their common SVP recursion. Segment endpoints returned by `SVP()` are one-based and inclusive; the last endpoint is always n .

Data model and simulation

Let μ_t be piecewise constant and

$$Y_t = \mu_t + E_t, \quad E_t = \rho E_{t-1} + \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad |\rho| < 1.$$

Equivalently, $Y_t = \mu_t + \rho(Y_{t-1} - \mu_{t-1}) + \eta_t$. The call `ts_generator(type = "gaussAR1", rho = rho, sd_noise = sigma)` implements this equation and initializes $Y_1 = \mu_1 + \eta_1/\sqrt{1-\rho^2}$. Thus `sd_noise` is the **innovation** standard deviation and the matching SVP argument is `sigma2 = sd_noise^2`; the stationary marginal variance is $\sigma^2/(1-\rho^2)$.

If `rho = NA_real_`, the package computes once, from the complete series,

$$M_k = \text{median}_{t=k+1, \dots, n} |Y_t - Y_{t-k}|, \quad \hat{\rho} = \text{clip}_{[-0.999, 0.999]} \left(\frac{M_2^2}{M_1^2} - 1 \right).$$

For stationary Gaussian AR(1) noise, $\text{Var}(Y_{t+k} - Y_t) \propto 1 - \rho^k$, so $M_2^2/M_1^2 = 1 + \rho$. Medians make `AR1_rho()` resistant to the relatively few differences crossing mean changes, although short or change-dense series can still give an imprecise estimate.

Exact conditional scan: AR1

Consider one candidate segment $x_1, \dots, x_m$ and condition on x_1 . Put $z_i = x_{i+1} - \rho x_i$, $i = 1, \dots, m-1$. Under no change, $z_i = (1-\rho)\mu + \eta_{i+1}$ and

$$RSS_0 = \min_{\mu} \sum_{i=1}^{m-1} \{z_i - (1-\rho)\mu\}^2 = \sum_{i=1}^{m-1} (z_i - \bar{z})^2.$$

For a change after observation τ , $2 \leq \tau \leq m-2$, write $\beta = (\mu_1, \mu_2)^\top$ and define the $(m-1) \times 2$ design

$$(X_\tau)_i = \begin{cases} (1-\rho, 0), & i < \tau, \\ (-\rho, 1), & i = \tau, \\ (0, 1-\rho), & i > \tau. \end{cases}$$

The middle row is essential: its residual is $x_{\tau+1} - \rho x_\tau - \mu_2 + \rho \mu_1$. Therefore

$$RSS_1(\tau) = \min_{\beta} \|z - X_\tau \beta\|^2, \quad RSS_1 = \min_{2 \leq \tau \leq m-2} RSS_1(\tau), \quad T_{\text{fix}} = \frac{RSS_0 - RSS_1}{2\sigma^2}.$$

`AR1_single_change()` returns these two RSS values, the minimizing τ , T_{fix} , and `valid = (T_fix < gamma)`. Prefix sums of z_i and z_i^2 make each $RSS_1(\tau)$ calculation $O(1)$ and the complete single-segment scan $O(m)$. Segments with fewer than four observations have statistic zero. `valid_AR1()` evaluates the same equations in R and returns only the validity indicator, for example for use as a test in `svp0()`.

Profiled scale and functional pruning

`test = "AR1Profile"` uses the same designs but maximizes separately over the innovation variance under the null and alternative:

$$T_{\text{prof}} = \frac{m-1}{2} \log \left(\frac{RSS_0}{RSS_1} \right).$$

It is exactly equivalent to `test = "AR1"`, `profile_sigma = TRUE`. `sigma2` must still be a positive argument but does not enter this statistic. Because T_{prof} and T_{fix} have different null laws, their values of `gamma` require separate calibration.

`test = "AR1Focus"` uses fixed ρ and σ^2 but avoids explicitly rescanning every τ . For each change location, the conditional log-likelihood is a concave quadratic in (μ_1, μ_2) . The C++ recurrence updates four functional envelopes (left/right and the two signs), retains only quadratic pieces that can be optimal, and includes the crossing-transition term above. If $\Delta_F(\tau)$ denotes the quadratic gain retained by this recurrence, its statistic is on the same half-likelihood-ratio scale:

$$T_{\text{Focus}} = \frac{1}{2\sigma^2} \max_{\tau} \Delta_F(\tau).$$

Although the internal class comment calls this recurrence exact, the user-facing help correctly treats it as an approximation: current numerical paths do not satisfy $T_{\text{Focus}} = T_{\text{fix}}$ for every segment. Therefore `AR1Focus` and `AR1` often agree, but endpoint equality is not guaranteed; this point is quantified on page 4.

The common SVP problem

For a segment $I_{s,t} = \{s+1, \dots, t\}$, let $V_{\gamma}(s, t)$ mean that the selected AR(1) statistic is strictly below γ . All three choices solve

$$\hat{\tau} \in \arg \text{lex min}_{\substack{0=\tau_0 < \dots < \tau_K=n \\ V_{\gamma}(\tau_{k-1}, \tau_k) \forall k}} \left(K, \sum_{k=1}^K C(\tau_{k-1}, \tau_k) \right), \quad C(s, t) = \sum_{i=s+1}^t (Y_i - \bar{Y}_{s+1:t})^2.$$

Thus the AR(1) likelihood determines **validity**, while ordinary within-segment SSE is the secondary tie-breaker in the current generic SVP implementation. With $R_t = (K_t, Q_t)$ and $R_0 = (0, 0)$,

$$R_t = \text{lex min}_{0 \leq s < t: V_{\gamma}(s, t)} \{R_s + (1, C(s, t))\}.$$

The exact test costs $O(m)$ whenever the statistic of a length- m candidate is requested. Consequently, without candidate pruning, the generic SVP loop has $O(n^2)$ candidate–endpoint pairs but an $O(n^3)$ worst-case total for `AR1` and `AR1Profile`. `AR1Focus` replaces the inner scan by functional updates and is generally faster; its work depends on how many quadratic pieces survive. `subtests = "right"`, `"left"`, or `"both"` removes SVP candidates; `"none"` is the assumption-free reference when the needed pruning implications have not been established for a chosen statistic.

Parameterization

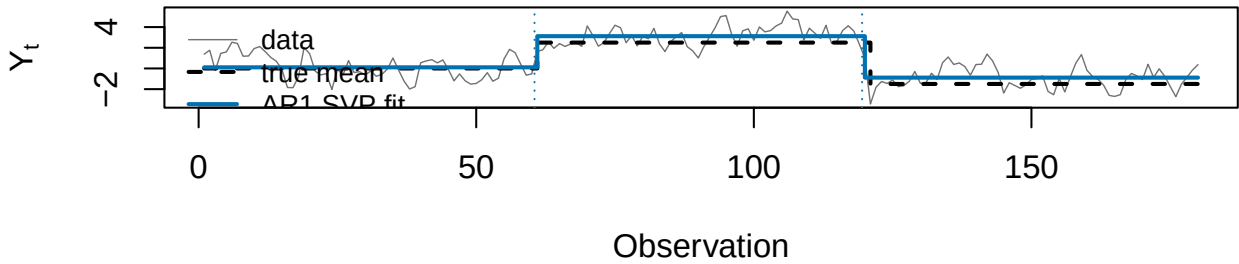
Purpose	Call and interpretation
fixed ρ, σ^2	<code>SVP(y, gamma, "AR1", rho=rho, sigma2=sigma^2)</code>
unknown σ^2	<code>SVP(y, gamma, "AR1Profile", rho=rho)</code>
same profiled test	<code>SVP(y, gamma, "AR1", rho=rho, profile_sigma=TRUE)</code>
functional recurrence	<code>SVP(y, gamma, "AR1Focus", rho=rho, sigma2=sigma^2)</code>
unknown ρ	use <code>rho=NA_real_</code> or first compute <code>AR1_rho(y)</code>
R validity wrapper	<code>valid_AR1(y, gamma, rho, sigma2, profile_sigma)</code>
validity pruning	use <code>subtests="none"</code> for the unpruned benchmark

One simulated series

The following reproducible example has $n = 180$, changes after observations 60 and 120, $\rho = 0.7$, $\sigma = 0.8$, and $\gamma = 6$. No constant-valued segments are used: every observation contains Gaussian AR(1) noise.

```
set.seed(20260914)
y <- ts_generator(c(60,120,180), c(0,2.5,-1.5),
                  sd_noise=.8, rho=.7, type="gaussAR1")
SVP(y, 6, test="AR1", subtests="none", rho=.7, sigma2=.8^2)
```

Method	rho used	inclusive segment endpoints
AR1	0.700	60, 119, 180
AR1Profile	0.700	60, 119, 180
AR1Focus	0.700	60, 119, 180
AR1; rho estimated	0.759	60, 119, 180



The exact, profiled, and functional implementations return the same three-segment partition here; estimating ρ gives $\hat{\rho} = 0.759$ and the same endpoints.

Repeated simulation

To check behavior beyond one realization, the experiment was repeated 100 times with the same model and threshold. The table reports the proportion with the correct three segments and the median Hausdorff distance d_H between estimated and true internal changepoints. A fit with no internal changepoint is penalized by n . These are finite-sample illustrations, not a calibration of `gamma`; fixed and profiled statistics should be calibrated separately for inferential comparisons.

Method	Pr(correct K=3)	median d_H
AR1	0.96	1
AR1Profile	0.96	1
AR1Focus	0.96	1
AR1; rho estimated	0.71	2

`AR1` and `AR1Focus` returned identical endpoint vectors in 100/100 repetitions. For the runs estimating ρ , $\text{mean}(\hat{\rho}) = 0.699$ and $\text{sd}(\hat{\rho}) = 0.240$. The loss in correct segment count illustrates why a known or independently estimated ρ can matter, even though the median localization error remains small in this example.

The simulations deliberately use `subtests = "none"`: they compare the three statistics without confounding them with validity-pruning assumptions. For a production analysis, calibrate `gamma` for the chosen statistic, variance treatment, pruning mode, and sample size.

Which results should be identical?

The required parameter matching is summarized below. Here, identical means the same returned endpoint vector, including n .

Comparison	Expected result
AR1Profile versus AR1 with <code>profile_sigma=TRUE</code>	Identical: both select the same C++ test class and profiled branch.
<code>rho=NA_real_</code> versus <code>rho=AR1_rho(y)</code> in the same test	Identical: the same full-series estimator is evaluated before SVP.
fixed-scale AR1 versus AR1Focus	Often identical, but not guaranteed: they use different numerical recurrences.
fixed-scale AR1 versus AR1Profile	Not expected: their statistics and threshold calibrations differ.

The 100 moderate-signal simulations above happened to give identical **AR1** and **AR1Focus** endpoints. A broader falsification check used $n \in \{40, 80, 160\}$, $\rho \in \{-0.8, -0.4, 0, 0.3, 0.7, 0.9\}$, $\gamma \in \{1.5, 3, 6, 10\}$ and 20 seeds: 32 of 1,440 endpoint vectors differed. This is numerical evidence that the equality is not universal, not a measure of the methods' statistical quality.

Time complexity and empirical confirmation

For **AR1** and **AR1Profile**, testing a length- m segment scans $O(m)$ change locations. With no SVP pruning, the total work is therefore

$$\sum_{t=1}^n \sum_{s=0}^{t-1} O(t-s) = O(n^3).$$

For **AR1Focus**, let $P_{s,t}$ be the number of quadratic pieces inspected when $I_{s,t}$ is extended. Its work is $O(\sum_{s < t} P_{s,t})$: the worst case remains $O(n^3)$ because $P_{s,t} \leq t-s$, but effective functional pruning typically keeps $P_{s,t}$ small and gives near- $O(n^2)$ behavior.

The following median elapsed times (seconds) use five runs, R 4.6.1 on arm64, the same noisy three-segment AR(1) model as above, and `subtests="none"` so all $n(n+1)/2$ candidate–endpoint pairs are identical across methods.

n	AR1	AR1Profile	AR1Focus
500	0.033	0.034	0.013
1000	0.272	0.274	0.056
2000	2.181	2.159	0.253

The log–log slopes are AR1 3.02, AR1Profile 2.99, AR1Focus 2.14. At $n = 2000$, **AR1Focus** is 8.6 times faster than **AR1**. Thus the expected ranking is confirmed for a controlled, large unpruned workload. With `left`, `right`, or `both`, wall time also depends on how many SVP candidates each statistic removes; timings are then not a pure comparison of the three segment-test algorithms.

Reference. Chakar, S., Lebarbier, E., L'evy-Leduc, C. and Robin, S. (2017). A robust approach for estimating change-points in the mean of an AR(1) process. *Bernoulli*, 23(2), 1408–1447.